

ACM India Summer School on Symmetric Key
Cryptography
Day 2: Negligible Functions, PRGs, and PRFs
Solutions

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1 Negligible Functions

A function $g(k)$ is negligible (denoted $g(k) \in \text{negl}(k)$) if for every constant $c > 0$, there exists a $k' \in \mathbb{N}$ such that for all $k \geq k'$:

$$|g(k)| < \frac{1}{k^c}$$

Claim 1. *Let negl_1 and negl_2 be two negligible functions. Then $g(k) = \text{negl}_1(k) + \text{negl}_2(k)$ is also negligible.*

Proof. We need to prove that for any arbitrary constant $c > 0$, there exists a k' such that for all $k \geq k'$, $\text{negl}_1(k) + \text{negl}_2(k) < \frac{1}{k^c}$.

Let $c' = c + 1$. Since negl_1 and negl_2 are negligible:

- There exists k_1 such that for all $k \geq k_1$, $\text{negl}_1(k) < \frac{1}{2k^{c'}}$ (by considering the polynomial $2k^c$).
- There exists k_2 such that for all $k \geq k_2$, $\text{negl}_2(k) < \frac{1}{2k^{c'}}$.

Let $k' = \max\{k_1, k_2\}$. For all $k \geq k'$:

$$\text{negl}_1(k) + \text{negl}_2(k) < \frac{1}{2k^{c'}} + \frac{1}{2k^{c'}} = \frac{1}{k^c}$$

Thus, the sum is negligible. $\square$

Claim 2. *A negligible function stays negligible even after polynomial multiplication: $\text{poly}(k) \cdot \text{negl}(k) = \text{negl}(k)$.*

Proof. Let $p(k) = k^a$ for some constant $a > 0$ represent the polynomial, and $g(k)$ be a negligible function. We want to show that $p(k) \cdot g(k)$ is negligible. For

any constant $c > 0$, let $c' = c + a$. Since $g(k)$ is negligible, for the constant c' , there exists k' such that for all $k \geq k'$:

$$g(k) < \frac{1}{k^{c'}} = \frac{1}{k^{c+a}}$$

Multiplying both sides by $p(k) = k^a$:

$$p(k) \cdot g(k) < k^a \cdot \frac{1}{k^{c+a}} = \frac{1}{k^c}$$

Thus, $p(k) \cdot g(k)$ is negligible. $\square$

Claim 3. *If $f(k)$ is not a negligible function and $g(k)$ is a negligible function, then $h(k) = f(k) - g(k)$ is not a negligible function.*

Proof. We prove this by contradiction. Assume that $h(k) = f(k) - g(k)$ is a negligible function. Since the sum of two negligible functions is negligible (by Claim 1), and both $h(k)$ and $g(k)$ are negligible under our assumption, it follows that:

$$f(k) = h(k) + g(k)$$

must also be a negligible function. This directly contradicts the given assumption that $f(k)$ is not a negligible function. Therefore, the assumption that $f(k) - g(k)$ is negligible must be false. Hence, $f(k) - g(k)$ is not negligible. $\square$

2 Pseudorandom Generators (PRGs)

Claim 4. *Let $G_1, G_2 : \{0, 1\}^n \rightarrow \{0, 1\}^{2n}$ be two secure PRGs with $G_1 \neq G_2$. Define $G(s) = G_1(s) \oplus G_2(s)$. Is G necessarily a PRG?*

Proof. No, G is not necessarily a PRG. We show a counterexample. Let $G_1 : \{0, 1\}^n \rightarrow \{0, 1\}^{2n}$ be a secure PRG. Define $G_2(s) = G_1(s) \oplus 1^{2n}$. Since G_1 is a secure PRG, its XOR with a constant string 1^{2n} is also a secure PRG. However, computing $G(s)$ gives:

$$G(s) = G_1(s) \oplus G_2(s) = G_1(s) \oplus G_1(s) \oplus 1^{2n} = 1^{2n}$$

The output of $G(s)$ is always the constant string 1^{2n} . A distinguisher $D(w)$ that outputs 1 if $w = 1^{2n}$ and 0 otherwise distinguishes $G(s)$ from the uniform distribution U_{2n} with probability $1 - 2^{-2n}$, which is non-negligible. Thus, G is not a secure PRG. $\square$

Claim 5. *Let G be a secure PRG with expansion factor $\ell(n) > 2n$. Say whether G_1 is necessarily a PRG:*

1. $G_1(s) = G(s_1 \dots s_{\lfloor n/2 \rfloor})$
2. $G_1(s) = G(0^{|s|} \parallel s)$

3. $G_1(s) = G(s) \parallel G(s+1)$

Solutions.

1. **No.** G_1 does not satisfy the expansion requirement. If $\ell(n) = 2n + 1$, then for the input of length n , the output of $G_1(s)$ has length $\ell(\lfloor n/2 \rfloor) = 2\lfloor n/2 \rfloor + 1 \leq n + 1$, which is not strictly greater than the input size n for all cases (e.g. for odd n).
2. **No.** We can construct a counterexample. Let $\hat{G} : \{0, 1\}^{2n} \rightarrow \{0, 1\}^{\ell(2n)}$ be a secure PRG. Define $G(s_1)$ as:

$$G(s_1) = \begin{cases} 0^{\ell(2n)} & \text{if the first } n \text{ bits of } s_1 \text{ are all zeros} \\ \hat{G}(s_1) & \text{otherwise} \end{cases}$$

Since the probability of a uniform s_1 starting with 0^n is 2^{-n} (which is negligible), G is still a secure PRG. But if we define $G_1(s) = G(0^{|s|} \parallel s)$, the first half of its seed is always $0^{|s|}$, meaning $G_1(s)$ always outputs $0^{\ell(2n)}$. This is trivially distinguishable from a random distribution.

3. **No.** We show a counterexample. Let $G' : \{0, 1\}^n \rightarrow \{0, 1\}^{\ell(n)-1}$ be a secure PRG. We construct a secure PRG $G : \{0, 1\}^n \rightarrow \{0, 1\}^{\ell(n)}$ as follows:

$$G(s) = \begin{cases} G'(s) \parallel 0 & \text{if } s \text{ is even (i.e. the LSB of } s \text{ is 0)} \\ G'(s-1) \parallel 1 & \text{if } s \text{ is odd (i.e. the LSB of } s \text{ is 1)} \end{cases}$$

Since LSB of a uniform seed s is 0 or 1 with probability $1/2$, the last bit of $G(s)$ is uniform, and $G(s)$ is a secure PRG.

Now, let's analyze $G_1(s) = G(s) \parallel G(s+1)$ where $s+1$ is addition modulo 2^n :

- If s is even, $s+1$ is odd. Then:

$$G_1(s) = [G'(s) \parallel 0] \parallel [G'((s+1)-1) \parallel 1] = G'(s) \parallel 0 \parallel G'(s) \parallel 1$$

Here, the last bit of the first half is 0, and the last bit of the second half is 1.

- If s is odd, $s+1$ is even. Then:

$$G_1(s) = [G'(s-1) \parallel 1] \parallel [G'(s+1) \parallel 0]$$

Here, the last bit of the first half is 1, and the last bit of the second half is 0.

In both cases, the last bit of the first half of $G_1(s)$ and the last bit of the second half are always different (i.e. their XOR is always 1). A distinguisher can check if the XOR of these two bits is 1. For $G_1(s)$ it is always 1 (probability 1), whereas for a uniform random string it is 1 with probability $1/2$. This yields a distinguishing advantage of $1/2$, proving G_1 is not a secure PRG.

□

3 Pseudorandom Functions (PRFs)

Claim 6. Consider $F_{A,b}(x) = Ax + b \pmod{2}$, where A is an $n \times n$ boolean matrix and b is an n -bit vector. Show that F is not a secure PRF.

Proof. A secure PRF is computationally indistinguishable from a truly random function. However, the construction $F_{A,b}(x) = Ax + b \pmod{2}$ is linear (affine). It satisfies the following relation for any inputs $x, y \in \{0, 1\}^n$:

$$\begin{aligned} F_{A,b}(x) \oplus F_{A,b}(y) \oplus F_{A,b}(x \oplus y) &= (Ax + b) \oplus (Ay + b) \oplus (A(x \oplus y) + b) \\ &= Ax \oplus b \oplus Ay \oplus b \oplus Ax \oplus Ay \oplus b = b \end{aligned}$$

Thus, the XOR sum $F_{A,b}(x) \oplus F_{A,b}(y) \oplus F_{A,b}(x \oplus y)$ is always constant and equals the vector b for all x, y .

We construct a distinguisher D that queries the oracle O on:

1. u, v and computes $t_1 = O(u) \oplus O(v) \oplus O(u \oplus v)$.
2. x, y (distinct from u, v) and computes $t_2 = O(x) \oplus O(y) \oplus O(x \oplus y)$.

If $t_1 = t_2$, the distinguisher outputs 1 (guessing O is $F_{A,b}$); otherwise it outputs 0.

- If $O = F_{A,b}$, then $t_1 = t_2 = b$ always, so D outputs 1 with probability 1.
- If O is a truly random function, t_1 and t_2 are independent uniform random n -bit strings, so $t_1 = t_2$ with probability 2^{-n} .

The distinguishing advantage is $1 - 2^{-n}$, which is non-negligible. Thus, $F_{A,b}$ is not a secure PRF. $\square$

Claim 7. Let F be a length-preserving pseudorandom function. Show that the keyed function $F' : \{0, 1\}^{n-1} \rightarrow \{0, 1\}^{2n}$ defined by $F'_k(x) = F_k(0 \parallel x) \parallel F_k(1 \parallel x)$ is a pseudorandom function.

Proof. We prove security of F' using a hybrid argument. We assume for contradiction that there exists a PPT distinguisher D that distinguishes $F'_k(x)$ from a truly random function $R : \{0, 1\}^{n-1} \rightarrow \{0, 1\}^{2n}$ with non-negligible advantage.

Define the following hybrid distributions for a fixed input x :

- $H_0 = F_k(0 \parallel x) \parallel F_k(1 \parallel x)$ (which is the output of $F'_k(x)$)
- $H_1 = U_n \parallel F_k(1 \parallel x)$ (where the first half is replaced by a uniform n -bit random string U_n)
- $H_2 = U_n \parallel U_n$ (which is a uniformly random $2n$ -bit string)

If D distinguishes H_0 from H_2 with non-negligible advantage, then by the hybrid argument, D must distinguish H_0 from H_1 , or H_1 from H_2 (or both) with non-negligible advantage.

Case 1: Distinguishing H_0 from H_1 . Suppose $|\Pr[D(H_0) = 1] - \Pr[D(H_1) = 1]|$ is non-negligible. We construct a PPT distinguisher D_1 against the PRF F . Given oracle access to a function $\mathcal{O}$ (which is either F_k or a truly random function), D_1 does the following:

1. Query $\mathcal{O}$ on $0 \parallel x$ to obtain y .
2. Independently compute $F_k(1 \parallel x)$.
3. Query D on the input $y \parallel F_k(1 \parallel x)$.
4. Output whatever D outputs.

If $\mathcal{O} = F_k$, the input to D is distributed exactly as H_0 . If $\mathcal{O}$ is a truly random function, the input to D is distributed as H_1 . Thus, D_1 distinguishes F from a random function with non-negligible advantage, contradicting the security of F as a PRF.

Case 2: Distinguishing H_1 from H_2 . Suppose $|\Pr[D(H_1) = 1] - \Pr[D(H_2) = 1]|$ is non-negligible. We construct a PPT distinguisher D_2 against the PRF F similarly. Given oracle access to $\mathcal{O}$:

1. Query $\mathcal{O}$ on $1 \parallel x$ to obtain y .
2. Choose a uniform random string $r \leftarrow \{0, 1\}^n$.
3. Query D on the input $r \parallel y$.
4. Output whatever D outputs.

If $\mathcal{O} = F_k$, the input to D is distributed exactly as H_1 . If $\mathcal{O}$ is a truly random function, the input to D is distributed as H_2 . Thus, D_2 distinguishes F from a random function with non-negligible advantage, contradicting the security of F .

Since both cases lead to contradictions, the transition from H_0 to H_2 cannot be distinguished with non-negligible advantage. Therefore, $H_0 \approx_c H_2$, and $F'_k(x)$ is a pseudorandom function. $\square$